

TRIPOLI INTL, Libyan Arab Jamahiriya

WMO: 620100

Lat: 32.664N

Lon: 13.159E

Elev: 263

StdP: 14.56

Time Zone: 2.00 (E02)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.2	42.5	24.5	18.8	64.5	28.1	22.3	62.5	25.8	58.3	23.1	60.1	5.4	210	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.3	107.7	72.8	104.0	72.1	100.4	71.4	80.2	95.4	78.0	91.2	76.4	89.1	11.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	140.5	86.3	74.7	131.2	84.8	73.0	123.6	83.2	44.0	95.3	41.6	90.8	39.9	88.7	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.6	21.3	18.9	35.1	113.9	2.7	2.4	33.2	115.7	31.6	117.1	30.1	118.4	28.2	120.2		
(5)			32.7	83.5	2.4	5.1	31.0	87.2	29.6	90.2	28.3	93.1	26.5	96.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	70.4	55.2	55.7	61.3	67.6	74.9	80.9	83.8	85.4	82.0	75.0	64.9	57.2	(6)
(7)		DBStd	12.44	4.05	5.42	6.77	7.35	7.56	7.07	5.55	4.98	5.47	5.80	5.64	4.76	(7)
(8)		HDD50	13	4	7	0	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD65	1105	304	267	159	50	4	0	0	0	0	1	70	250	(9)
(10)		CDD50	7460	166	167	350	528	771	927	1049	1097	960	774	447	224	(10)
(11)		CDD65	3079	2	7	43	128	311	477	584	632	510	310	67	8	(11)
(12)		CDH74	35921	24	109	536	1425	3495	5802	7485	8167	5618	2675	508	77	(12)
(13)	CDH80	19422	3	30	212	651	1849	3341	4391	4762	2868	1160	137	18	(13)	
(14)	Wind	WSAvg	8.3	7.4	8.1	8.4	9.7	9.9	9.6	8.1	8.0	8.4	7.3	7.2	7.7	(14)
(15)	Precipitation	PrecAvg	10.8	2.3	1.2	1.0	0.7	0.1	0.0	0.0	0.4	1.3	1.7	2.1	(15)	
(16)		PrecMax	20.7	6.8	3.3	3.9	4.1	1.0	0.7	0.2	0.1	5.4	6.8	8.3	(16)	
(17)		PrecMin	1.9	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)	
(18)		PrecStd	4.1	1.7	0.9	1.0	0.9	0.2	0.1	0.0	0.7	1.5	1.6	2.1	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.1	84.2	94.5	100.1	107.4	112.2	111.5	109.6	106.6	100.0	89.1	82.0	(19)
MCWB			55.7	58.6	64.3	68.2	71.1	75.3	72.8	73.4	73.5	69.9	64.2	57.7	(20)	
2%		DB	71.5	76.7	85.8	93.0	100.8	106.3	106.4	105.7	102.0	94.8	84.1	74.9	(21)	
		MCWB	55.1	57.3	62.1	63.8	69.6	71.0	72.7	74.0	70.6	68.5	62.6	58.4	(22)	
5%		DB	67.7	71.4	79.1	87.7	95.5	100.9	101.9	102.0	97.2	91.0	80.2	70.0	(23)	
		MCWB	54.6	55.5	59.2	63.6	67.9	71.4	72.1	73.0	71.3	67.4	62.2	56.3	(24)	
10%		DB	64.3	66.6	74.6	82.6	91.0	96.1	97.1	98.2	93.3	86.4	76.3	66.5	(25)	
		MCWB	53.4	54.5	58.3	61.7	66.2	69.6	70.9	72.5	71.2	67.5	61.1	55.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.3	63.3	69.6	75.3	79.3	84.0	81.2	83.4	81.3	77.2	70.4	64.4	(27)
MCD B			68.7	74.4	83.0	87.8	98.9	106.7	99.5	99.1	93.6	88.5	80.9	71.9	(28)	
2%		WB	57.6	59.4	64.4	68.7	74.2	78.1	77.6	80.0	78.3	73.8	67.2	60.7	(29)	
		MCD B	65.8	71.5	78.5	84.5	93.0	97.1	92.9	93.3	88.8	84.2	76.5	68.8	(30)	
5%		WB	56.1	57.3	61.6	65.4	70.6	74.3	76.0	77.9	76.4	71.9	65.2	58.7	(31)	
		MCD B	63.7	67.9	74.1	81.3	87.7	90.0	90.4	90.0	86.6	82.5	74.4	66.3	(32)	
10%		WB	54.9	55.5	59.6	63.0	68.2	72.1	74.5	76.3	74.8	70.2	63.3	57.0	(33)	
		MCD B	62.2	64.2	71.0	77.9	83.9	86.1	88.1	88.3	85.5	80.7	72.2	64.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.2	20.1	22.4	23.3	24.0	24.1	23.9	23.3	21.0	20.7	19.8	17.9	(35)
MCDBR			23.7	28.4	30.0	31.1	30.3	30.5	29.6	28.3	26.3	26.2	24.5	23.7	(36)	
MCWBR			12.9	14.2	13.7	13.0	11.8	11.9	11.5	10.7	9.6	10.6	11.2	13.0	(37)	
5% WB		MCDBR	19.5	24.3	26.1	27.3	26.9	25.6	24.2	23.3	21.4	21.8	21.0	19.3	(38)	
		MCWBR	12.4	14.5	14.0	14.1	13.1	12.3	11.4	11.0	9.9	10.9	12.0	12.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.380	0.404	0.481	0.537	0.539	0.485	0.486	0.484	0.515	0.472	0.420	0.388	(40)
taud		2.220	2.137	1.922	1.814	1.836	1.992	2.004	2.024	1.947	2.025	2.118	2.196	(41)		
Ebn at Noon		259	265	252	244	245	257	256	255	240	239	241	248	(42)		
Edh at Noon		36	43	58	68	67	57	56	54	56	48	40	35	(43)		
(44)	All-Sky Solar Radiation	RadAvg		982	1258	1663	2015	2268	2473	2508	2259	1825	1375	1055	856	(44)
(45)		RadStd		61	70	97	110	97	95	45	59	73	85	47	65	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page